WEEK 6: SAREE BOUTIQUE REVIEW AND RATING MANAGEMENT SYSTEM

Scenario

The Saree Boutique wants to collect customer feedback about purchased sarees. Customers can provide ratings and reviews after completing their orders. The system should store customer opinions, analyze saree performance, and identify highly rated sarees.

The Saree Boutique Review and Rating Module is connected with the existing Customer, Order and Product modules.

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1. CREATE REVIEW AND RATING TABLE

Objective

To create a table for storing customer reviews and ratings for purchased sarees.

```
SQL> CREATE TABLE REVIEW (
  2      REVIEW_ID VARCHAR2(7) PRIMARY KEY,
  3      CUSTOMER_ID VARCHAR2(7),
  4      PRODUCT_ID VARCHAR2(7),
  5      RATING NUMBER(1),
  6      REVIEW_TEXT VARCHAR2(50),
  7      REVIEW_DATE DATE
  8  );

Table created.

SQL> INSERT INTO REVIEW VALUES ('REV0001','CUS0001','PROD001',5,'Excellent product',SYSDATE);

1 row created.
```

The screenshot where the complete CREATE TABLE SAREE_REVIEW code and the output **"Table created."** appear in the same image.

---

2. STORE CUSTOMER RATINGS FROM 1 TO 5 STARS

Objective

To ensure that customers can provide ratings only from 1 to 5 stars.

```
SQL> ALTER TABLE SAREE_REVIEW ADD CONSTRAINT CHK_SAREE_RATING CHECK (RATING BETWEEN 1 AND 5);

Table altered.
```

The screenshot where the complete ALTER TABLE SAREE_REVIEW ADD CONSTRAINT CHK_SAREE_RATING... code and **"Table altered."** appear together.

---

3. ADD REVIEWS FOR PURCHASED SAREES

Objective

To insert customer reviews and ratings for purchased sarees.

```
SQL> INSERT INTO SAREE_REVIEW
  2  (REVIEW_ID, ORDER_ID, CUSTOMER_ID, PRODUCT_ID, RATING, REVIEW_COMMENT, REVIEW_DATE)
  3  VALUES
  4  ('REV0001', 'ORD0001', 'CUS0001', 'P001', 5, 'Excellent saree quality and beautiful design', DATE '2026-09-24');

1 row created.

SQL> INSERT INTO SAREE_REVIEW
  2  (REVIEW_ID, ORDER_ID, CUSTOMER_ID, PRODUCT_ID, RATING, REVIEW_COMMENT, REVIEW_DATE)
  3  VALUES
  4  ('REV0002', 'ORD0002', 'CUS0002', 'P002', 4, 'Good quality saree with attractive colour', DATE '2026-09-24');

1 row created.

SQL> INSERT INTO SAREE_REVIEW
  2  (REVIEW_ID, ORDER_ID, CUSTOMER_ID, PRODUCT_ID, RATING, REVIEW_COMMENT, REVIEW_DATE)
  3  VALUES
  4  ('REV0003', 'ORD0001', 'CUS0001', 'P101', 5, 'Very elegant saree and worth the price', DATE '2026-09-25');

1 row created.

SQL> INSERT INTO SAREE_REVIEW
  2  (REVIEW_ID, ORDER_ID, CUSTOMER_ID, PRODUCT_ID, RATING, REVIEW_COMMENT, REVIEW_DATE)
  3  VALUES
  4  ('REV0004', 'ORD0002', 'CUS0002', 'P102', 3, 'Average quality but nice design', DATE '2026-09-26');

1 row created.

SQL> INSERT INTO SAREE_REVIEW
  2  (REVIEW_ID, ORDER_ID, CUSTOMER_ID, PRODUCT_ID, RATING, REVIEW_COMMENT, REVIEW_DATE)
  3  VALUES
  4  ('REV0005', 'ORD0001', 'CUS0001', 'P111', 2, 'Design is good but quality needs improvement', DATE '2026-09-27');

1 row created.
```

Save the Records

```
SQL> COMMIT;

Commit complete.
```

The screenshot showing the review insert command and successful **“1 row created.”** output.
If you have multiple insert screenshots, add all the screenshots showing the successful insertions.

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4. DISPLAY ALL SAREE REVIEWS

Objective

To display all reviews stored in the Saree Boutique database.

```
SQL> SELECT REVIEW_ID, ORDER_ID, CUSTOMER_ID, PRODUCT_ID,
  2         RATING, REVIEW_COMMENT, REVIEW_DATE
  3  FROM SAREE_REVIEW;

REVIEW_ ORDER_I CUSTOME PRODUCT     RATING
------- ------- ------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------
REV0001 ORD0001 CUS0001 P001             5
Excellent saree quality and beautiful design
24-SEP-26

REV0002 ORD0002 CUS0002 P002             4
Good quality saree with attractive colour
24-SEP-26

REVIEW_ ORDER_I CUSTOME PRODUCT     RATING
------- ------- ------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------

REV0003 ORD0001 CUS0001 P101             5
Very elegant saree and worth the price
25-SEP-26

REV0004 ORD0002 CUS0002 P102             3
Average quality but nice design

REVIEW_ ORDER_I CUSTOME PRODUCT     RATING
------- ------- ------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------
26-SEP-26

REV0005 ORD0001 CUS0001 P111             2
Design is good but quality needs improvement
27-SEP-26
```

The screenshot where the above SELECT query and the complete review result table are visible in the same image.

---

5. UPDATE REVIEW COMMENTS

Objective

To update a review comment when required.

```
SQL> UPDATE SAREE_REVIEW
  2  SET REVIEW_COMMENT = 'Excellent saree quality, beautiful design and comfortable to wear'
  3  WHERE REVIEW_ID = 'REV0001';

1 row updated.
```

The screenshot showing the UPDATE command and **“1 row updated.”**

Verify the Updated Review

```
SQL> SELECT REVIEW_ID, PRODUCT_ID, RATING, REVIEW_COMMENT
  2  FROM SAREE_REVIEW
  3  WHERE REVIEW_ID = 'REV0001';

REVIEW_ PRODUCT     RATING
------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REV0001 P001             5
Excellent saree quality, beautiful design and comfortable to wear
```

The screenshot where the verification query and the updated review comment are visible together.

---

6. REMOVE INAPPROPRIATE OR INVALID REVIEWS

Objective

To remove an invalid or inappropriate review from the system.

```
SQL> DELETE FROM SAREE_REVIEW
  2  WHERE REVIEW_ID = 'REV0005';

1 row deleted.
```

The screenshot showing the DELETE command and **"1 row deleted."**

```
SQL> COMMIT;

Commit complete.
```

The screenshot showing the COMMIT; command and successful commit output, if displayed.

---

7. RETRIEVE SAREE REVIEW DETAILS

7.1 Display All Reviews for a Saree

```
SQL> SELECT PRODUCT_ID, REVIEW_ID, RATING, REVIEW_COMMENT, REVIEW_DATE
  2  FROM SAREE_REVIEW
  3  WHERE PRODUCT_ID = 'P001';

PRODUCT REVIEW_     RATING
------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------
P001    REV0001          5
Excellent saree quality, beautiful design and comfortable to wear
24-SEP-26
```

The screenshot where the above query and all reviews for P001 are displayed together.

7.2 Display Customer Name with Their Reviews



```
SQL> SELECT O.CUSTOMER_NAME,
  2         R.PRODUCT_ID,
  3         R.RATING,
  4         R.REVIEW_COMMENT,
  5         R.REVIEW_DATE
  6  FROM SAREE_REVIEW R
  7  JOIN ORDERS O
  8  ON R.ORDER_ID = O.ORDER_ID;

CUSTOMER_NAME                                      PRODUCT     RATING
-------------------------------------------------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------
                                                   P001             5
Excellent saree quality, beautiful design and comfortable to wear
24-SEP-26

                                                   P002             4
Good quality saree with attractive colour
24-SEP-26

CUSTOMER_NAME                                      PRODUCT     RATING
-------------------------------------------------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------

                                                   P101             5
Very elegant saree and worth the price
25-SEP-26

                                                   P102             3
Average quality but nice design

CUSTOMER_NAME                                      PRODUCT     RATING
-------------------------------------------------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------
26-SEP-26
```

The screenshot where the query and the result showing **CUSTOMER_NAME + PRODUCT_ID + RATING + REVIEW_COMMENT** appear together.

7.3 Find Sarees Having Maximum Reviews

Objective

To identify the sarees that have received the highest number of customer reviews.

```
SQL> SELECT PRODUCT_ID, COUNT(*) AS TOTAL_REVIEWS
  2  FROM SAREE_REVIEW
  3  GROUP BY PRODUCT_ID
  4  HAVING COUNT(*) = (
  5      SELECT MAX(REVIEW_COUNT)
  6      FROM (
  7          SELECT COUNT(*) AS REVIEW_COUNT
  8          FROM SAREE_REVIEW
  9          GROUP BY PRODUCT_ID
 10      )
 11  );

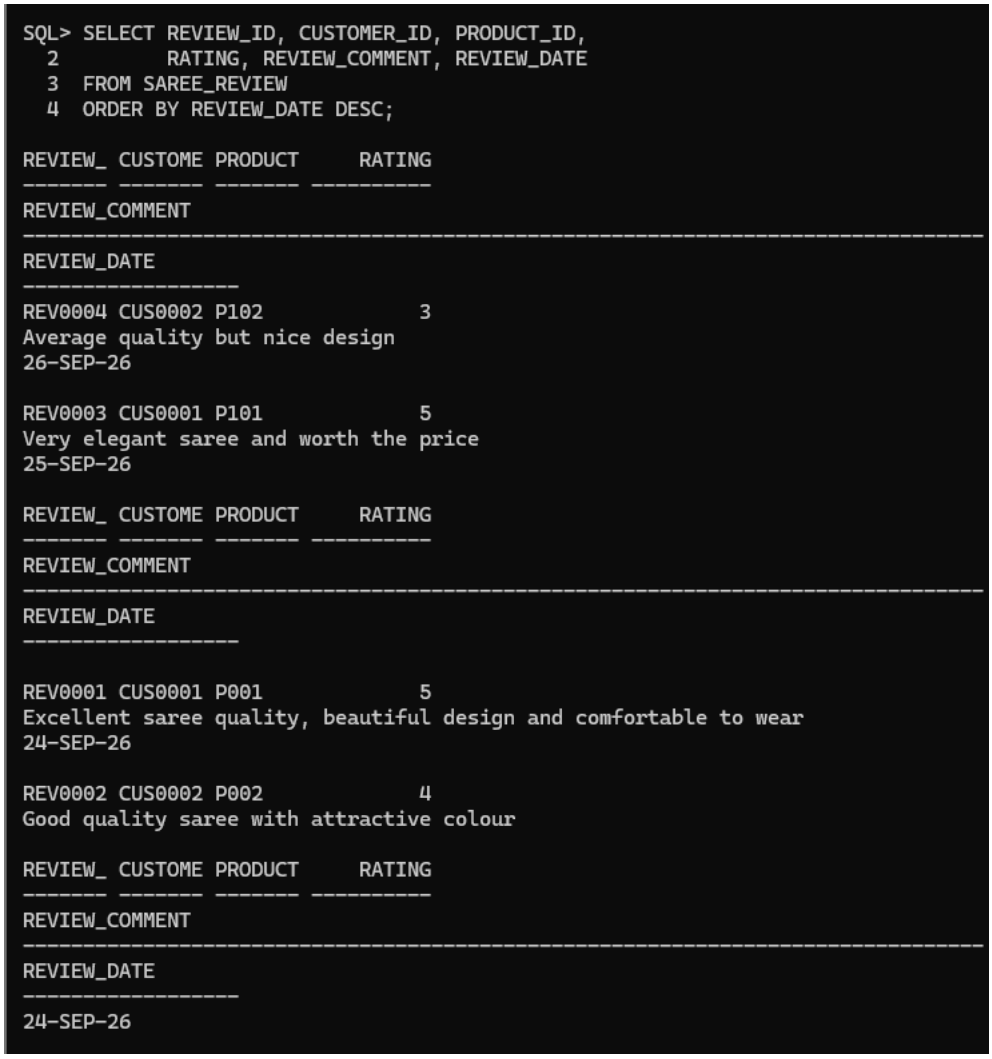
PRODUCT TOTAL_REVIEWS
------- -------------
P001                1
P102                1
P101                1
P002                1
```

The screenshot where the complete query and the saree(s) having the maximum number of reviews are displayed.

7.4 Display Recent Customer Feedback

Objective

To display the most recent customer feedback first.



The screenshot where the query and the reviews arranged from the latest date to the oldest date are visible.

7.5 Retrieve Reviews with Ratings Above 4

Objective

To display highly rated saree reviews.

```
SQL> SELECT REVIEW_ID, PRODUCT_ID, RATING,
  2         REVIEW_COMMENT, REVIEW_DATE
  3  FROM SAREE_REVIEW
  4  WHERE RATING > 4;

REVIEW_ PRODUCT     RATING
------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------
REV0001 P001             5
Excellent saree quality, beautiful design and comfortable to wear
24-SEP-26

REV0003 P101             5
Very elegant saree and worth the price
25-SEP-26

REVIEW_ PRODUCT     RATING
------- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DATE
------------------
```

The screenshot where the query and only reviews having ratings **5** are displayed.

---

8. CALCULATE AVERAGE SAREE RATINGS

Objective

To analyze the ratings given by customers using aggregate functions.

8.1 Calculate Average Rating for Each Saree

```
SQL> SELECT PRODUCT_ID,
  2         ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  3  FROM SAREE_REVIEW
  4  GROUP BY PRODUCT_ID;

PRODUCT AVERAGE_RATING
------- --------------
P001                 5
P102                 3
P101                 5
P002                 4
```

The screenshot where the query and each PRODUCT_ID with its AVERAGE_RATING are visible.

8.2 Count Number of Reviews per Saree

```
SQL> SELECT PRODUCT_ID,
  2         COUNT(*) AS NUMBER_OF_REVIEWS
  3  FROM SAREE_REVIEW
  4  GROUP BY PRODUCT_ID;

PRODUCT NUMBER_OF_REVIEWS
------- -----------------
P001                  1
P102                  1
P101                  1
P002                  1
```

The screenshot where the query and the number of reviews for each saree are displayed.

8.3 Find Highest-Rated Sarees

Objective

To identify the sarees having the highest average rating.

```
SQL> SELECT PRODUCT_ID,
  2         ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  3  FROM SAREE_REVIEW
  4  GROUP BY PRODUCT_ID
  5  HAVING AVG(RATING) = (
  6      SELECT MAX(AVG_RATING)
  7      FROM (
  8          SELECT AVG(RATING) AS AVG_RATING
  9          FROM SAREE_REVIEW
 10          GROUP BY PRODUCT_ID
 11      )
 12  );

PRODUCT AVERAGE_RATING
------- --------------
P001                5
P101                5
```

The screenshot where the query and the highest-rated saree(s) are displayed.

8.4 Identify Sarees with Average Rating Above 4

Objective

To identify sarees that have received an average rating greater than 4.

```
SQL> SELECT PRODUCT_ID,
  2         ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  3  FROM SAREE_REVIEW
  4  GROUP BY PRODUCT_ID
  5  HAVING AVG(RATING) > 4;

PRODUCT AVERAGE_RATING
------- --------------
P001                5
P101                5
```

The screenshot where the query and all sarees having an average rating above 4 are visible.

---

9. IDENTIFY HIGHLY RATED SAREES

Objective

To generate reports based on saree ratings and customer feedback.

REPORT 1: SAREE RATING ANALYSIS

Report Purpose

This report displays the saree ID, number of customer reviews and average rating.

```
SQL> SELECT PRODUCT_ID AS SAREE_ID,
  2         COUNT(*) AS NUMBER_OF_REVIEWS,
  3         ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  4  FROM SAREE_REVIEW
  5  GROUP BY PRODUCT_ID
  6  ORDER BY AVERAGE_RATING DESC;

SAREE_I NUMBER_OF_REVIEWS AVERAGE_RATING
------- ----------------- --------------
P001                    1              5
P101                    1              5
P002                    1              4
P102                    1              3
```

The screenshot where the complete query and the final **Saree Rating Analysis** result are visible together.

Report Contains

- Saree ID
- Number of Reviews
- Average Rating

REPORT 2: CUSTOMER FEEDBACK ANALYSIS

Report Purpose

This report identifies the most reviewed sarees, highly rated sarees and sarees requiring improvement.

9.1 Most Reviewed Sarees

```
SQL> SELECT PRODUCT_ID AS SAREE_ID,
  2         COUNT(*) AS NUMBER_OF_REVIEWS
  3  FROM SAREE_REVIEW
  4  GROUP BY PRODUCT_ID
  5  ORDER BY NUMBER_OF_REVIEWS DESC;

SAREE_I NUMBER_OF_REVIEWS
------- -----------------
P001                    1
P002                    1
P101                    1
P102                    1
```

The screenshot where the query and the sarees arranged according to the number of reviews are displayed.

9.2 Highly Rated Sarees

```
SQL> SELECT PRODUCT_ID AS SAREE_ID,
  2         ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  3  FROM SAREE_REVIEW
  4  GROUP BY PRODUCT_ID
  5  HAVING AVG(RATING) > 4
  6  ORDER BY AVERAGE_RATING DESC;

SAREE_I AVERAGE_RATING
------- --------------
P101                 5
P001                 5
```

The screenshot where the query and highly rated sarees with average rating above 4 are displayed.

9.3 Sarees Requiring Improvement

```
SQL> SELECT PRODUCT_ID AS SAREE_ID,
  2         ROUND(AVG(RATING), 2) AS AVERAGE_RATING
  3  FROM SAREE_REVIEW
  4  GROUP BY PRODUCT_ID
  5  HAVING AVG(RATING) < 3
  6  ORDER BY AVERAGE_RATING;

no rows selected
```

The screenshot where the query and sarees having average rating below 3 are displayed.

REPORT 3: RATING DISTRIBUTION

Report Purpose

This report shows the distribution of customer ratings for the Saree Boutique.

10.1 Number of 5-Star Ratings

```
SQL> SELECT COUNT(*) AS FIVE_STAR_RATINGS
  2  FROM SAREE_REVIEW
  3  WHERE RATING = 5;

FIVE_STAR_RATINGS
-----------------
                2
```

The screenshot where the query and the number of 5-star ratings are visible.

10.2 Number of 4-Star Ratings

```
SQL> SELECT COUNT(*) AS FOUR_STAR_RATINGS
  2  FROM SAREE_REVIEW
  3  WHERE RATING = 4;

FOUR_STAR_RATINGS
-----------------
                1
```

The screenshot where the query and the number of 4-star ratings are visible.

10.3 Number of Low-Rated Sarees

```
SQL> SELECT COUNT(*) AS LOW_RATED_REVIEWS
  2  FROM SAREE_REVIEW
  3  WHERE RATING <= 2;

LOW_RATED_REVIEWS
-----------------
                0
```

The screenshot where the query and the number of low-rated reviews are visible.

10.4 Complete Rating Distribution

```
SQL> SELECT RATING,
  2         COUNT(*) AS NUMBER_OF_RATINGS
  3  FROM SAREE_REVIEW
  4  GROUP BY RATING
  5  ORDER BY RATING DESC;

    RATING NUMBER_OF_RATINGS
---------- -----------------
         5                 2
         4                 1
         3                 1
```

The screenshot where the complete rating distribution from **5 to 1 stars** is displayed.

---

11. FINAL SAREE RATING SUMMARY

Objective

To provide a complete summary of customer ratings for the Saree Boutique.

```
SQL> SELECT RATING,
  2         COUNT(*) AS NUMBER_OF_CUSTOMERS
  3  FROM SAREE_REVIEW
  4  GROUP BY RATING
  5  ORDER BY RATING DESC;

    RATING NUMBER_OF_CUSTOMERS
---------- -------------------
         5                   2
         4                   1
         3                   1
```

The screenshot showing the final rating summary with rating values and number of customers.

CONCLUSION

The Saree Boutique Review and Rating Management System successfully stores customer reviews and ratings for purchased sarees. The system allows the boutique to update and remove reviews, retrieve customer feedback, calculate average ratings, identify highly rated sarees, find sarees requiring improvement and analyze the overall rating distribution.

The generated reports help the Saree Boutique understand customer satisfaction and improve saree quality and services based on customer feedback.